

AERO40007 – Mechanics

Imperial College of Science, Technology & Medicine

Department of Aeronautics

1. A force $F = 10i + 3j - 2k$ N goes through a point whose position vector $R = 6i - 2j$ m. Find an equivalent system such that the force goes through the point $R_1 = 2i + 3k$ m.

ANS. $F = 10i + 3j - 2k$ N and $M = 13i - 22j + 32k$ Nm

~~$$\vec{F}\vec{R} + \vec{M}\vec{R}_1 = 0$$~~

~~$$\therefore \vec{F} \times \vec{R} = \begin{vmatrix} i & j & k \\ 10 & 3 & -2 \\ 6 & -2 & 0 \end{vmatrix} = (3 \times 0 - (-2) \times (-2))i + (-2 \times 6 - 10 \times 0)j + (10 \times (-2) - 3 \times 6)k$$~~
~~$$= -4i - 12j - 38k$$~~

~~$$\therefore \vec{M}\vec{R}_1 = -4i - 12j - 38k$$~~

~~$$= \begin{vmatrix} i & j & k \\ a & b & c \\ 2 & 0 & 3 \end{vmatrix} = (3b - 0)c + (-2c - 3a)b + (0 - 2b)a$$~~

$$\vec{M} = \vec{R} \times \vec{F}$$

$$= (\vec{R} - \vec{R}_1) \times \vec{F}$$

$$= (4i - 2j - 3k) \times (10i + 3j - 2k)$$

$$= \begin{vmatrix} i & j & k \\ 4 & -2 & -3 \\ 10 & 3 & -2 \end{vmatrix} = (4 + 9)i + (-30 + 8)j + (12 + 20)k$$

$$= 13i - 22j + 32k \text{ (Nm)}$$

2. Three forces and two couples are given as follows:

$F_1 = (4i + 3j)$ N and acts through $(1,0,1)$ m

$F_2 = (5i + 5j + 5k)$ N and acts through $(1,1,1)$ m

$F_3 = (2j + 4k)$ N and acts through $(0,1,2)$ m

$M_1 = (6i + 6j + 6k)$ Nm

$M_2 = (3i - 3k)$ Nm

Replace them by a single force and moment acting through the origin.

ANS. $F = 9i + 10j + 9k$ N and $M = 6i + 10j + 6k$ Nm

$$F = F_1 + F_2 + F_3 = 9i + 10j + 9k \text{ (N)}$$

$$M = \vec{R}_1 \times \vec{F}_1 + \vec{R}_2 \times \vec{F}_2 + \vec{R}_3 \times \vec{F}_3 + M_1 + M_2$$

$$= \begin{vmatrix} i & j & k \\ 1 & 0 & 1 \\ 4 & 3 & 0 \end{vmatrix} + \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 5 & 5 & 5 \end{vmatrix} + \begin{vmatrix} i & j & k \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{vmatrix} + M_1 + M_2$$

$$= (-3i + 4j + 3k) + (6i + 6j + 6k) + (3i - 3k)$$

$$= 6i + 10j + 6k \text{ (Nm)}$$

3. What is the equivalent force system acting at point A in Fig 1?

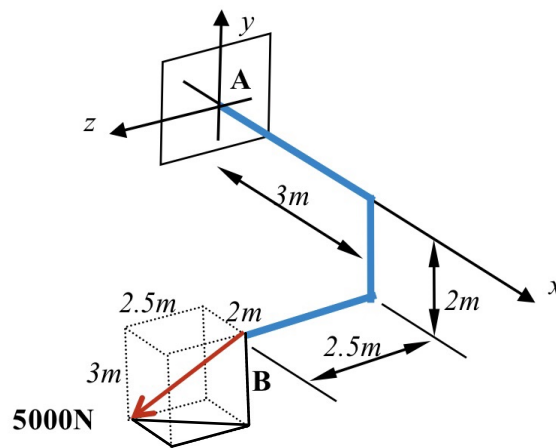


Figure 1.

ANS. $F = -2279i - 3419j + 2849k$ N and $M = 2849i - 14245j - 14815k$ Nm

$$F_x = -5000 \text{ N} \times \frac{2}{\sqrt{(2.5)^2 + 2^2 + 3^2}} = -2279.21$$

$$F_y = -5000 \text{ N} \times \frac{3}{\sqrt{(2.5)^2 + 2^2 + 3^2}} = -3418.82$$

$$F_z = 5000 \text{ N} \times \frac{2.5}{\sqrt{(2.5)^2 + 2^2 + 3^2}} = 2849.01$$

$$\therefore F = -2279.21i - 3418.82j + 2849.01k \text{ (N)}$$

$$\therefore R = 3i - 2j + 2.5k \text{ (m)}$$

$$\therefore M = \vec{R} \times \vec{F} = \begin{vmatrix} i & j & k \\ 3 & -2 & 2.5 \\ F_x & F_y & F_z \end{vmatrix} = 2849.03i - 14245.06j - 14814.88k$$